



## AQ2.3: Activity Questions 3 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1:

1 point

Match the matrices in Column A with their inverses in Column B.

☐

a → → iii

☐

a → → ii

☐

b → → iii

☐

b → → i

☐

c → → i

☐

c → → ii

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

a → → ii

b → → iii

c → → i

**(Use the below information to answer questions 2 and 3).**

Consider a system of linear equations

$$\begin{aligned} 2x_1 - x_2 &= 3 \\ x_1 - x_3 &= 3 \\ x_2 - x_3 &= 2 \end{aligned}$$

Let the matrix representation of the above system be  $Ax = b$ , where  $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$  and  $b = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$ .

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \text{ and } b = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}.$$

1 point

Choose the set of correct options.

☐

$$A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}^{-1}$$

☐


$$A^{-1} = \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix} A^{-1} = \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix}$$

☐

$$\text{Adjoint of the matrix } AA \text{ is } \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix} \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix}$$

☐

$$\det(A) = 1 \det(A) = 1.$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$A^{-1} = \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix} A^{-1} = \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix}$$

$$\text{Adjoint of the matrix } AA \text{ is } \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix} \begin{bmatrix} 1 & -11 \\ 1 & -22 \\ 1 & -21 \end{bmatrix}$$

$$\det(A) = 1 \det(A) = 1.$$

1 point

Choose the set of correct options.

☐

$$x_1 = -2x_1 = -2.$$

☐

$$x_2 = 1x_2 = 1.$$

☐

$$x_3 = -1x_3 = -1.$$

☐ None of the above.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$x_2 = 1x_2 = 1.$$

$$x_3 = -1x_3 = -1.$$

1 point

Choose the set of correct options.

☐

A system of linear equations  $Ax = b$  is called a homogeneous system of linear equations if  $b \neq 0$ .

☐

A system of linear equations  $Ax = b$  is called a non-homogeneous system of linear equations if  $b \neq 0$ .

☐

If  $v$  is a solution of the system of linear equations  $Ax = b$ , then  $cv$  is a solution of system of linear equations  $cAx = b$ , where  $c \neq 0$ .

☐

Let  $AX = b$  be a system of linear equations. If  $A$  is invertible, then  $adj(A)x = b$  also has a solution.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

A system of linear equations  $AX = b$  is called a non-homogeneous system of linear equations if  $b \neq 0$ .

If  $v$  is a solution of the system of linear equations  $AX = b$ , then  $cv$  is a solution of system of linear equations  $cAX = cb$ , where  $c \neq 0$ .

Let  $AX = b$  be a system of linear equations. If  $A$  is invertible, then  $adj(A)x = b$  also has a solution.

Suppose  $C$  is the cofactor matrix of  $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$ . Let  $C_{ij}$  be the  $ij$ -th element of the cofactor matrix  $C$ . Answer the questions 5,6 and 7.

Find the value of  $C_{12}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

Find the value of  $C_{22}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) -3

1 point

Find the sum of the elements in the third row of  $C$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

**Level 2:**

1 point

Which of the following square matrices of order 3 are the same as their adjoint matrices?

- ☐ Identity matrix.
- ☐ Zero matrix.
- ☐ Any scalar matrix.

☐ Any diagonal matrix.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Identity matrix.

Zero matrix.

1 point

Choose the set of correct options.

[Hint: For the first option, find out the adjoint matrix of an arbitrary square upper triangular matrix of order 22, i.e., take the matrix  $\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$  and find out its adjoint matrix.]

☐

The adjoint of a  $2 \times 22 \times 2$  real upper triangular matrix is an upper triangular matrix.

☐

There exists a square matrix of order 33, such that  $A = A^{-1} = \text{adj}(A)A = A^{-1} = \text{adj}(A)$ .

☐

If  $A = A^{-1}A = A^{-1}$ , then  $\det(A)\det(A)$  must be 11.

☐

If  $A = A^{-1}A = A^{-1}$ , then  $AA$  must be an identity matrix.

☐

If  $A = A^{-1}A = A^{-1}$ , then  $A^2 = IA2 = I$ .

☐

If  $A^{-1} = \text{adj}(A)A^{-1} = \text{adj}(A)$ , then  $\det(A)\det(A)$  must be 11.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The adjoint of a  $2 \times 22 \times 2$  real upper triangular matrix is an upper triangular matrix.

There exists a square matrix of order 33, such that  $A = A^{-1} = \text{adj}(A)A = A^{-1} = \text{adj}(A)$ .

If  $A = A^{-1}A = A^{-1}$ , then  $A^2 = IA2 = I$ .

If  $A^{-1} = \text{adj}(A)A^{-1} = \text{adj}(A)$ , then  $\det(A)\det(A)$  must be 11.

1 point

If  $A$  and  $BA$  and  $B$  are squares matrices of order 22, then choose the set of correct options.

[Hint: Find out the adjoint matrix of an arbitrary square matrix of order 22, i.e., take the matrix  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  and find out its adjoint matrix. ]

☐

$\text{adj}(AB) = \text{adj}(A)\text{adj}(B)\text{adj}(AB) = \text{adj}(A)\text{adj}(B)$ .

☐

$\text{adj}(AB) = \text{adj}(B)\text{adj}(A)\text{adj}(AB) = \text{adj}(B)\text{adj}(A)$ .

☐

$\text{adj}(A + B) = \text{adj}(A) + \text{adj}(B)\text{adj}(A + B) = \text{adj}(A) + \text{adj}(B)$ .

☐

$\text{adj}(A^T) = \text{adj}(A)^T \text{adj}(AT) = \text{adj}(A)^T$ .

☐

$\text{adj}(A^{-1}) = \text{adj}(A)^{-1} \text{adj}(A^{-1}) = \text{adj}(A)^{-1}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\text{adj}(AB) = \text{adj}(B)\text{adj}(A)\text{adj}(AB) = \text{adj}(B)\text{adj}(A)$ .

$\text{adj}(A + B) = \text{adj}(A) + \text{adj}(B)\text{adj}(A + B) = \text{adj}(A) + \text{adj}(B)$ .

$$\text{adj}(A^T) = \text{adj}(A)^T \quad \text{adj}(AT) = \text{adj}(A)^T.$$

$$\text{adj}(A^{-1}) = \text{adj}(A)^{-1} \quad \text{adj}(A^{-1}) = \text{adj}(A)^{-1}.$$

Check Answers

Your score is: 0/10